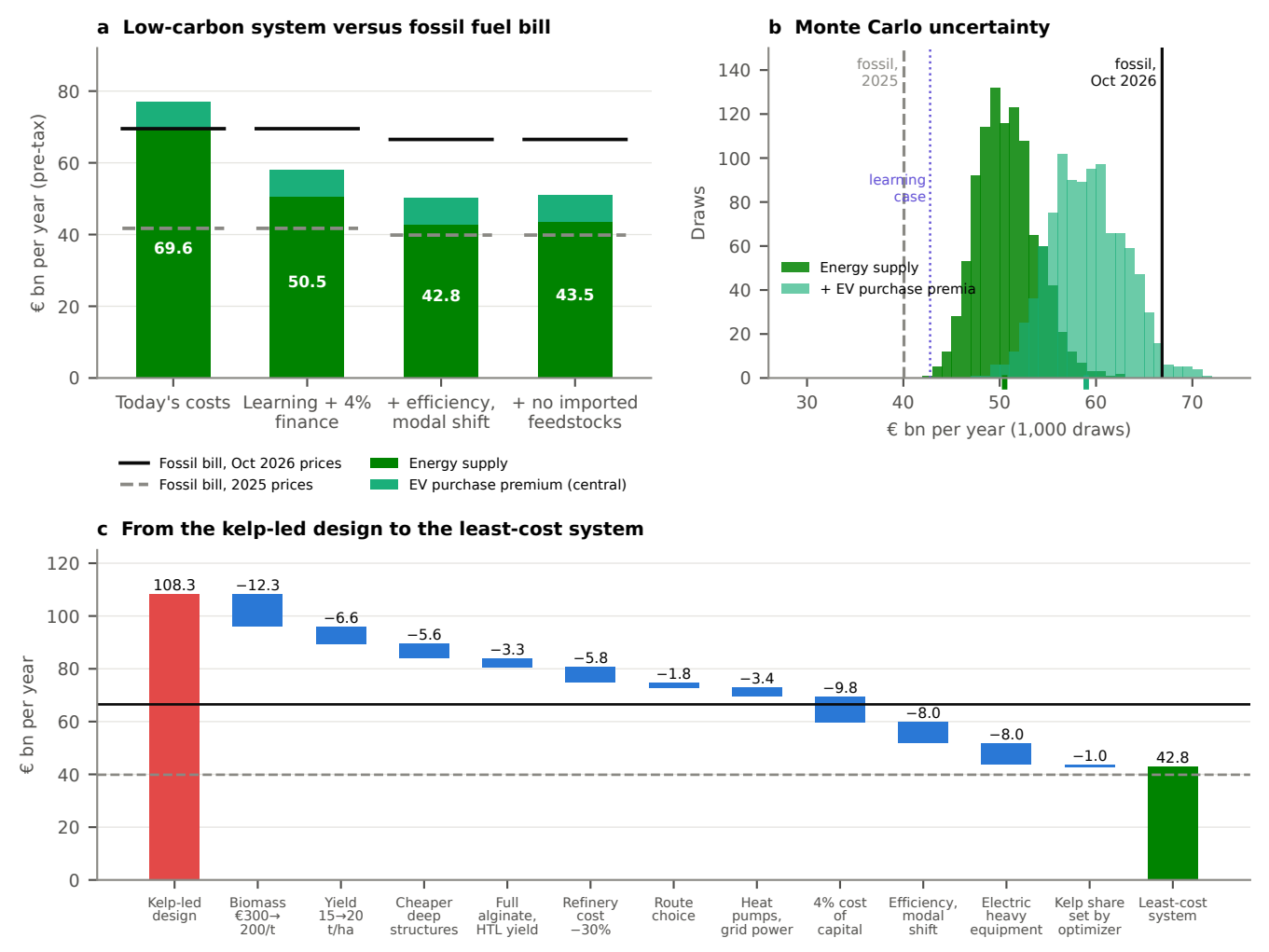
a Low-carbon system versus fossil fuel bill
€ bn per year (pre-tax)
0
20
40
60
80
69.6
50.5
42.8
43.5
Today's costs
Learning + 4% finance
+ efficiency, modal shift
+ no imported feedstocks
Fossil bill, Oct 2026 prices
Fossil bill, 2025 prices
Energy supply
EV purchase premium (central)
b Monte Carlo uncertainty
Draws
fossil, 2025
learning case
fossil, Oct 2026
Energy supply
+ EV purchase premia
30
40
50
60
70
€ bn per year (1,000 draws)
c From the kelp-led design to the least-cost system
€ bn per year
108.3
−12.3
−6.6
−5.6
−3.3
−5.8
−1.8
−3.4
−9.8
−8.0
−8.0
−1.0
42.8
Kelp-led design
Biomass €300→200/t
Yield 15→20 t/ha
Cheaper deep structures
Full alginate, HTL yield
Refinery cost −30%
Route choice
Heat pumps, grid power
4% cost of capital
Efficiency, modal shift
Electric heavy equipment
Kelp share set by optimizer
Least-cost system